

Note 14.2 Limits and Continuity in Higher Dimensions

Limits for Functions of Two Variables

Definition

We say that a function $f(x, y)$ approaches the **limit** L as (x, y) approaches (x_0, y_0) and write

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = L$$

if for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f ,

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta \implies |f(x, y) - L| < \epsilon$$

For a Single Variable

it can be shown that

$$\begin{aligned}\lim_{(x,y) \rightarrow (x_0,y_0)} x &= x_0 \\ \lim_{(x,y) \rightarrow (x_0,y_0)} y &= y_0 \\ \lim_{(x,y) \rightarrow (x_0,y_0)} k &= k \text{ (any number } k\text{)}\end{aligned}$$

Theorem 1--Properties of Limits of Functions of Two Variables

THEOREM 1—Properties of Limits of Functions of Two Variables

The fol-

lowing rules hold if L , M , and k are real numbers and

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

1. *Sum Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) + g(x,y)) = L + M$$

2. *Difference Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) - g(x,y)) = L - M$$

3. *Constant Multiple Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} kf(x,y) = kL \quad (\text{any number } k)$$

4. *Product Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \cdot g(x,y)) = L \cdot M$$

5. *Quotient Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}, \quad M \neq 0$$

6. *Power Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^n = L^n, \quad n \text{ a positive integer}$$

7. *Root Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n},$$

n a positive integer, and if n is even, we assume that $L > 0$.

Continuity

Definition

A function $f(x, y)$ is **continuous** at the point (x_0, y_0) if

- f is defined at (x_0, y_0)
- $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ exists.
- $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(x_0, y_0)$.

Two-Path Test for Nonexistence of a Limit

If a function $f(x, y)$ has **different limits along two different paths** in the domain of f as (x, y) approaches (x_0, y_0) , then $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ does not exist.

(But having the same limit along all straight lines approaching (x_0, y_0) does not imply the limit exists.)

Continuity of Composites

If f is continuous at (x_0, y_0) and g is a **single-variable** function continuous at $f(x_0, y_0)$, then the composite function $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is continuous at (x_0, y_0) .

Functions of More than Two Variables

The conclusions for functions of two variables can also extend to functions of three or more variables.

Extreme Values of Continuous Functions on Closed, Bounded Sets

For a function more than one variables, there still exists that if it is continuous on a closed, bounded set R in the plane then it has absolute minimum and maximum at some point in R .

Exercises

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